

Decision-making guide

Logistics

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1. Simulation platform introduction

1.1 Game Arena

The gaming arena interface comprises various decision tabs that includes Inbound logistics, ware house decisions, routes and technologies, outbound logistics, Decision Checklist, Report, and Synopsis. Some sections must be completed first as they have an impact on other areas.

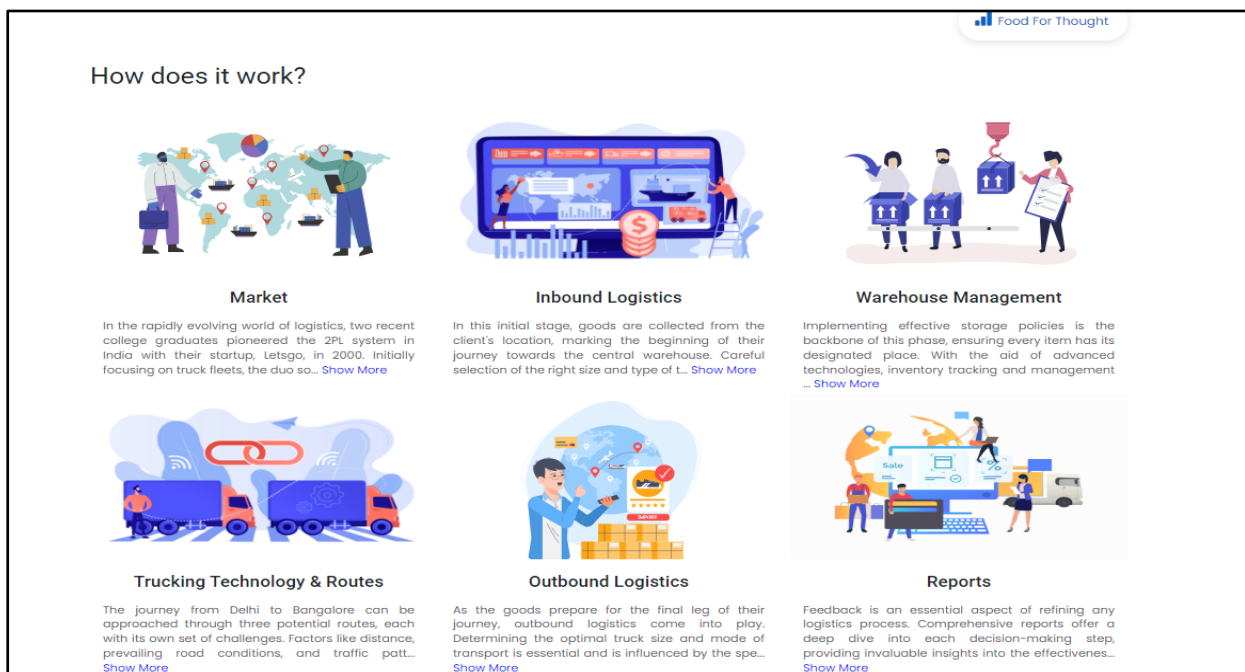


Figure 1 Game Arena

1.2 Readings

This section offers crucial materials that participants need to go through for active participation in the game. The primary reading material includes a decision-making guide. Instructors can choose to enhance these resources with additional materials. Additionally, there is a walkthrough video available, providing comprehensive details about the business scenario covered in the simulation. This video offers insights into the company, its industry, current trends, challenges, and the various decisions that need to be made.

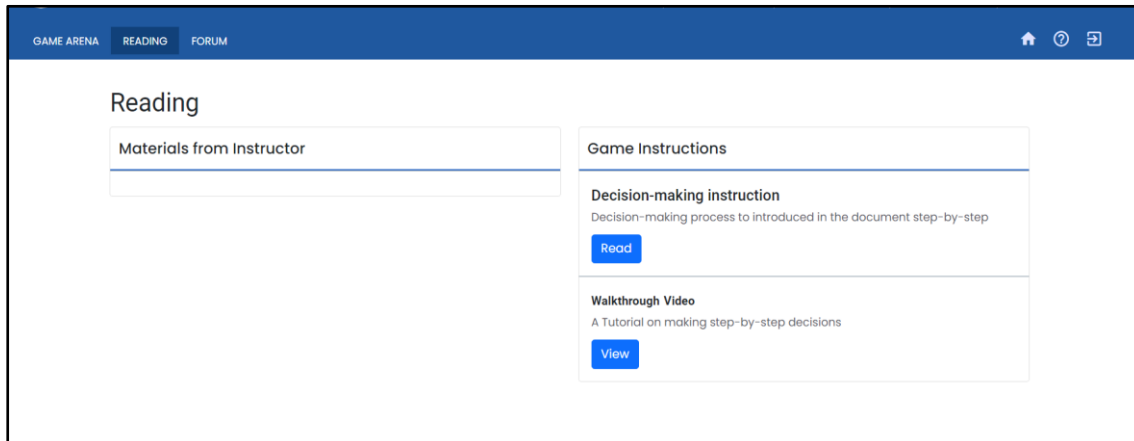


Figure 2 Reading Section

1.3 Forum

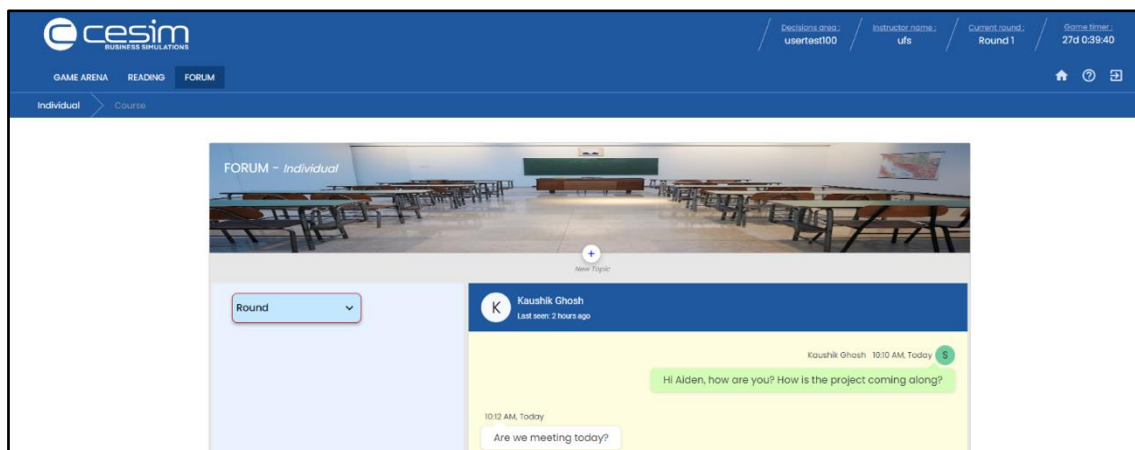


Figure 3 Forum Section

- The forums provide an excellent platform for participants to engage with their instructors or peers.
- There are two separate forums: the Individual Forum and the Course Forum. The Individual Forum allows participants to communicate directly with the instructor, while the Course Forum is accessible to all enrolled players.
- Instructors can view and respond to posts in both sections. Consequently, the Course Forum is an appropriate space for asking questions that can benefit the entire course, while the Individual Forum is the preferred channel for the individual to have a direct discussion with the professor.
- Participants will receive notifications for messages.

1.4 Main Objective & Winning Criteria

The simulation aims to cultivate a comprehensive perspective on overseeing both inbound and outbound logistics. It involves optimizing transportation methods and pathways while integrating technologies to reduce emissions and enhance effectiveness. Through this, participants will grasp the significance of meeting client needs and the value of timely order fulfilment.

1.5 Successful decision-making flow

The sequence of decision-making unfolds as follows:

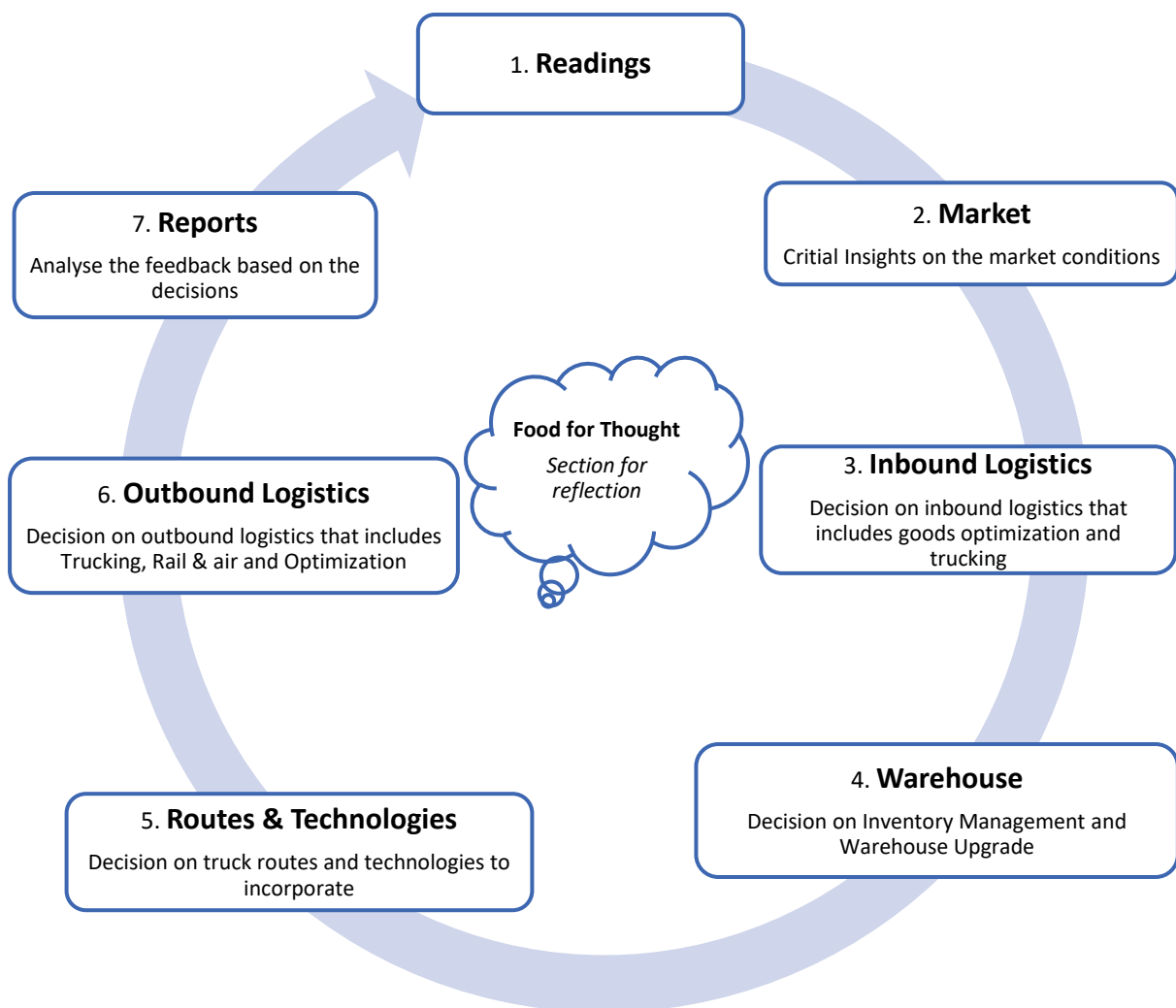


Figure 4 The sequence of decision making

2. Game Arena

2.1 Introduction

This page encompasses information detailing the functioning of the simulation and provides insights into each tab's decision-making.

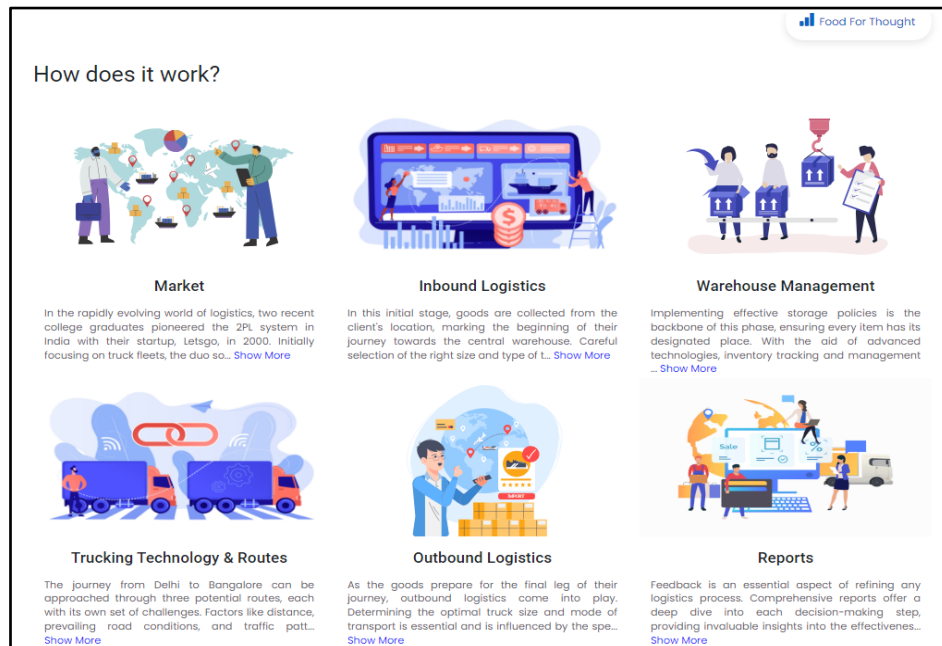


Figure 5 The Introduction Tab

2.2 Food for Thought

This segment provides crucial aspects to consider, offering valuable insights for decision-making.



Figure 6 The food for thought Section in the simulation

2.2 Market

Prior to commencing the decision-making process, it is essential to carefully examine the "Market" section. This section offers vital information on market trends, case scenarios, and key parameters to consider when making decisions in each tab.

Market	
Letsgo's logistics evolution is intricately shaped by four pivotal clients, each with unique needs. Unison Limited, a seasoned FMCG manufacturer and a longstanding partner, entrusts Letsgo with a diverse range of products, highlighting the company's commitment to reliability and excellence. Promton Incorporation, a leading player in FMCD manufacturing, relies on Letsgo for swift and efficient goods movement, particularly in the dynamic South Indian market, showcasing Letsgo's ability to meet specialized logistics requirements. Fix Corporate, a medium-sized textile manufacturer has found a dependable logistics partner in Letsgo. The textile industry's demands for precision and flexibility align with Letsgo's expertise in optimizing routes and managing warehouse operations, ensuring seamless deliveries for Fix Corporate's goods. Lastly, for eCom Ltd, the largest e-retailer specializing in perishable items, Letsgo's services are of utmost importance. The time-sensitive nature of their products necessitates a logistics solution that guarantees timely deliveries while preserving product quality, showcasing Letsgo's commitment to efficient and reliable transport. As the newly appointed Logistics Consultant, your critical responsibility is orchestrating the transportation of goods from these four key clients on their journey from Delhi to Bangalore. With these valued clients at the core of Letsgo's logistics landscape, your role becomes instrumental in the company's ongoing success.	
Parameters	
	This Period
Order, metric tonnes	
Unison Limited	875
Promton Incorporation	645
Fix Corporate	390
eCom Limited	612
Contractual TAT, days	
Unison Limited	8
Promton Incorporation	9
Fix Corporate	12
eCom Limited	5
Company Rate, INR per mt-km	
Small Trucks	3.6
Open Body Trucks	3.3

Figure 7 The Market Tab

2.3 Inbound Logistics

Participants are tasked with making decisions pertaining to inbound logistics, which involve considerations such as selecting the most suitable truck and optimizing goods transportation. The challenge lies in evaluating various options based on specific parameters tailored to each client, as well as factors like mileage, average speed, and minimizing damage during transportation. Participants must weigh these factors carefully to make informed decisions that align with the unique requirements and objectives of each client, while also maximizing efficiency and minimizing risks associated with the transportation process.

a. Trucking

When making decisions regarding trucking options for inbound logistics, participants must consider several factors, including the specific parameters under the market tab that are tailored to each of the four clients. These parameters include contractual turnaround time,

fragility contract terms, workforce requirements, and other relevant information related to inbound logistics.

By thoroughly analysing the food for thought questions and considering the specific parameters provided for each client, participants can make informed decisions regarding trucking options and the number of trucks to use for inbound logistics, ultimately optimizing efficiency and meeting client expectations effectively.

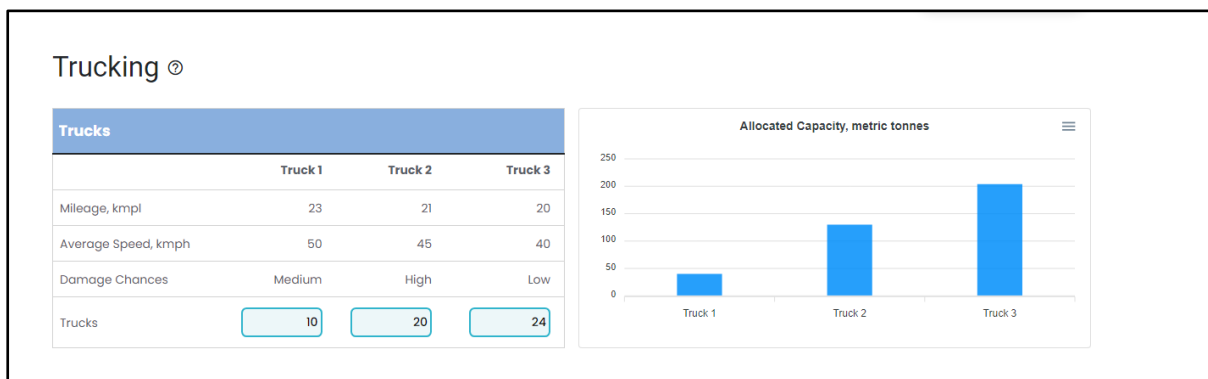
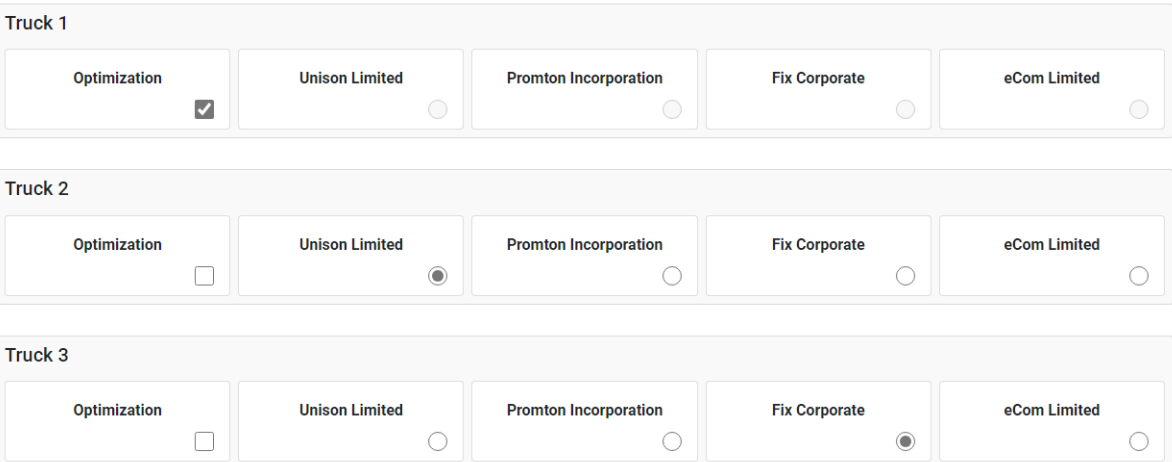


Figure 8 Decision-making regarding trucking options for inbound logistics

b. Goods Optimization

Here in this decision section, the participant have to take decision on goods optimization option, choose truck for specific clients' need. Factors to consider include fragility contract, orders of each client, workforce parameters, Contractual TAT days, and emission.



Goods Optimization

Truck 1

Optimization ☒ Unison Limited ☐ Promton Incorporation ☐ Fix Corporate ☐ eCom Limited ☐

Truck 2

Optimization ☐ Unison Limited ☒ Promton Incorporation ☐ Fix Corporate ☐ eCom Limited ☐

Truck 3

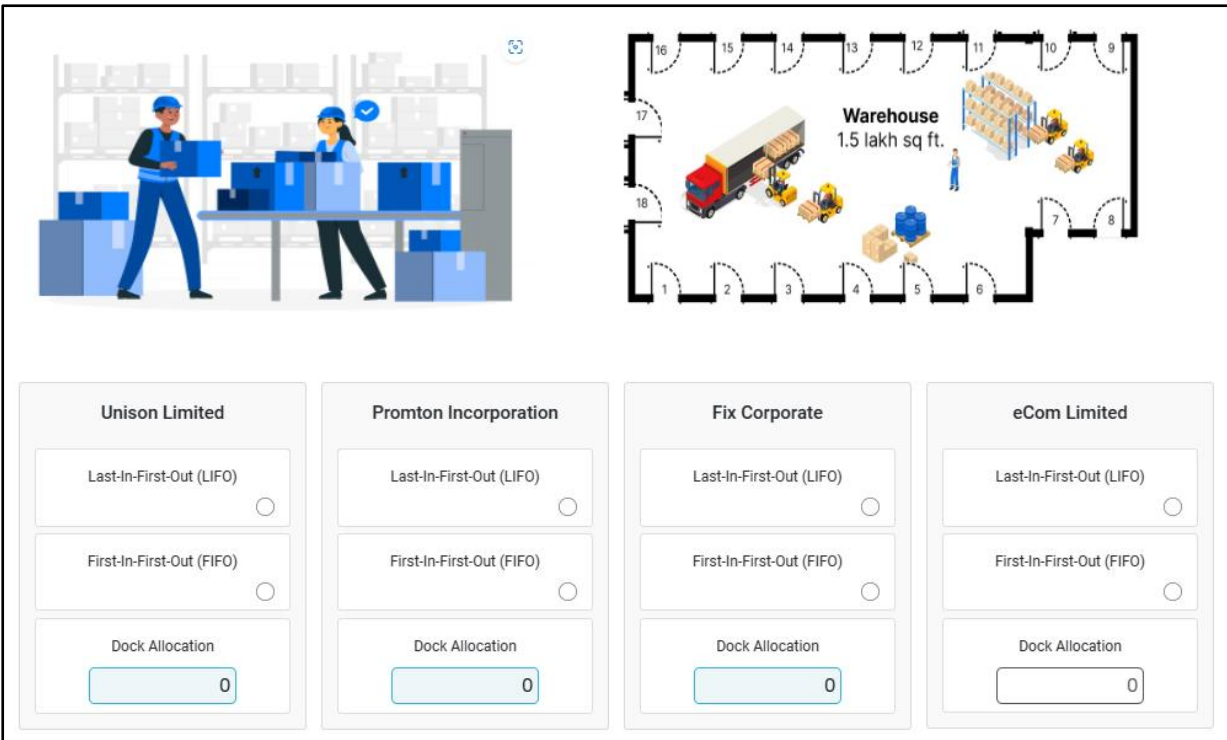
Optimization ☐ Unison Limited ☐ Promton Incorporation ☐ Fix Corporate ☒ eCom Limited ☐

Figure 9 Decision-making regarding goods optimization for inbound logistics

2.4 Warehouse

a. Inventory Management

Participants are tasked with selecting the appropriate inventory policy for each client company from two options: LIFO (Last In, First Out) and FIFO (First In, First Out) methods. Selecting the wrong inventory policy could disrupt the goods flow, leading to extended turnaround times that impact operational efficiency and customer satisfaction.



The screenshot displays the 'Warehouse' tab interface. At the top, there is an illustration of a warehouse with workers and a truck. Below this, a map of the warehouse layout is shown, labeled 'Warehouse 1.5 lakh sq ft.', with numbered zones from 1 to 18. The main part of the interface consists of four panels, one for each company:

Unison Limited	Promton Incorporation	Fix Corporate	eCom Limited
Last-In-First-Out (LIFO) ☐	Last-In-First-Out (LIFO) ☐	Last-In-First-Out (LIFO) ☐	Last-In-First-Out (LIFO) ☐
First-In-First-Out (FIFO) ☐	First-In-First-Out (FIFO) ☐	First-In-First-Out (FIFO) ☐	First-In-First-Out (FIFO) ☐
Dock Allocation 0	Dock Allocation 0	Dock Allocation 0	Dock Allocation 0

Figure 10 Inventory Management Decision in Warehouse Tab

b. Warehouse Upgrade

Choosing a warehouse upgrade aims to enhance efficiency, decrease energy consumption, and reduce emissions for an eco-friendlier facility. Each upgrade comes with a cost and offers advantages like increased efficiency, productivity, and process enhancement.

 Conveyor System It is a common mechanical equipment mainly used to move goods from one place to another. The system ... Show More Cost, INR 230000 ☐	 Stacker Cranes A machine designed for the automated storage of materials. They travel along the aisles of the wareh... Show More Cost, INR 280000 ☐	 Stretch Wrappers Stretch films are highly stretchable plastic which is wrapped around the boxes in order to keep the ... Show More Cost, INR 220000 ☐
 Warehouse Management System A technology solution that automates inventory control and syncs the different processes taking plac... Show More Cost, INR 380000 ☐	 Shuttle System The Shuttle System is an automated storage solution for boxes that speeds up order picking, ensuring... Show More Cost, INR 400000 ☐	

Figure 11 Warehouse Upgrade Decision in Warehouse Tab

2.5 Routes & Technologies

a. Routes

Participants here in this section, selects the best route from Delhi to Bangalore, considering factors like distance, road conditions, traffic, truck stops, demand, fuel costs, and safety. Each route presents unique challenges, making the decision complex yet crucial for efficient logistics.

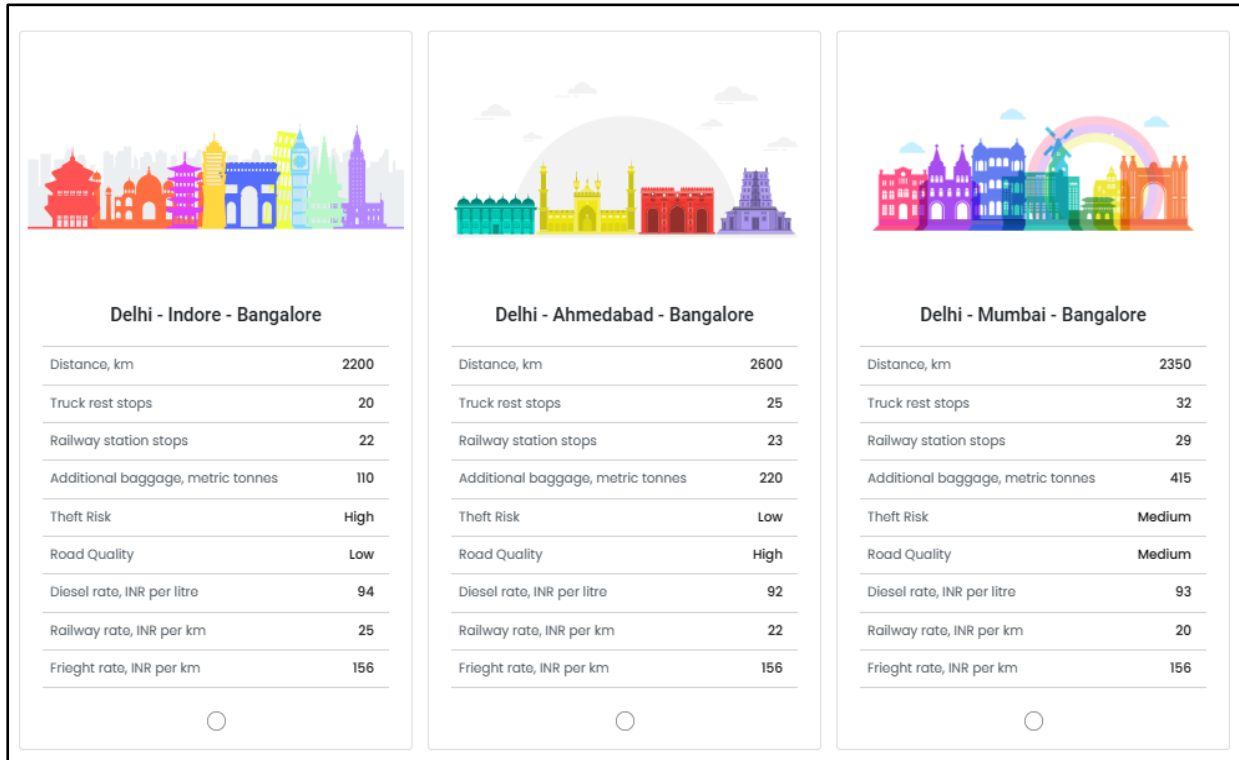


Figure 12 Routes Decision

b. Technology

The decision here includes the choice of right technology that aims at improving overall fuel efficiency, driver productivity, error and theft reduction etc. Each option's cost and advantages are mentioned in simulation.

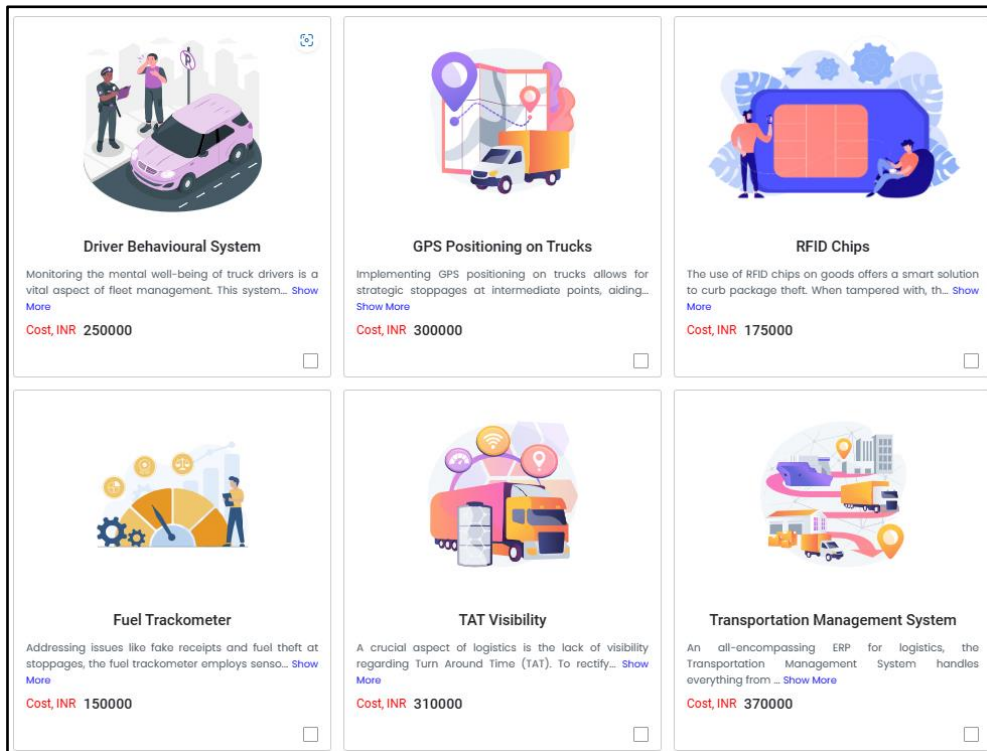


Figure 13 Technology Choices

2.6 Outbound Logistics

Participants optimize outbound logistics by selecting between truck, rail, and air transportation, with a focus on sustainability.

a. Trucks

Participants can opt for one of three truck varieties: small trucks, open-body trucks, and covered containers. When making decisions, factors such as fuel efficiency, average speed, risk of damage, and sustainability should be taken into consideration.

b. Rail & Air

Participants are tasked with selecting between rail and air transportation for outbound logistics, considering factors such as load capacity, lead time, average speed, fuel efficiency, and emissions.

Trucks			
	Small Trucks	Open Body Trucks	Covered Container
Mileage, kmpl	18	17	14
Average Speed, kmph	55	50	45
Damage Chances	Medium	High	Low
Trucks	0	0	0

Rail & Air Trips		
	Rail	Air
Lead Time	2	3
Average Speed, kmph	58	750
Damage Chances	High	Low
Number of Trips	0	0

Figure 14 Decisions on Mode of Transport in Outbound Logistics

c. Optimization

Participants are required to determine whether to choose bundling options and select the suitable truck type for specific clients or groups. Factors to take into account include contract fragility, client orders, workforce parameters, contractual TAT days, and emissions.

Small Trucks				
Optimization ☐	Unison Limited ☐	Prompton Incorporation ☐	Fix Corporate ☐	eCom Limited ☐
Open Body Trucks				
Optimization ☐	Unison Limited ☐	Prompton Incorporation ☐	Fix Corporate ☐	eCom Limited ☐
Covered Container				
Optimization ☐	Unison Limited ☐	Prompton Incorporation ☐	Fix Corporate ☐	eCom Limited ☐
Rail				
Optimization ☐	Unison Limited ☐	Prompton Incorporation ☐	Fix Corporate ☐	eCom Limited ☐
Air				
Optimization ☐	Unison Limited ☐	Prompton Incorporation ☐	Fix Corporate ☐	eCom Limited ☐

Figure 15 Optimization decision in Outbound Logistics

2.7 Decision checklist

The Decision Checklist assists participants in reviewing and modifying their decisions before the deadline by consolidating them from each round. Once decisions are finalized, participants can submit them using the Submit option if they opt to proceed with the Decision Checklist.

Decision Checklist

Round 1 

Participant Demo

Inbound	
Truck 1	30
Truck 2	20
Truck 3	30
Truck 1, Optimization	Yes
Truck 2, Optimization	eCom Limited
Truck 3, Optimization	Yes
Warehouse Management	
Unison Limited	First-In-First-Out (FIFO)
Prompton Incorporation	First-In-First-Out (FIFO)
Fix Corporate	First-In-First-Out (FIFO)
eCom Limited	First-In-First-Out (FIFO)
Unison Limited, docks	6
Prompton Incorporation, docks	5
Fix Corporate, docks	4
eCom Limited, docks	3
Warehouse Upgrade 1	Conveyor System
Warehouse Upgrade 2	Stacker Cranes
Routes & Technology	
Route	Delhi - Ahmedabad - Bangalore
Technology 1	GPS Positioning on Trucks
Technology 2	TAT Visibility
Outbound	
Small Trucks	40
Open Body Trucks	40
Covered Container	50
Rail Trips	4
Air Trips	3
Small Trucks, Optimization	Yes
Open Body Trucks, Optimization	Yes
Covered Container, Optimization	Yes
Rail, Optimization	Fix Corporate
Air, Optimization	eCom Limited

Submit

Figure 16 The decision checklist

2.8 Report

- The report section provides the details on the participants' critical thinking abilities in each aspect. It demonstrates how their thought process influenced the outcomes, providing insights into the impact of their decision-making skills.

This section consists of feedback on the performance of the participant based on four metrics.

- **Rigour:** Consistency throughout the decision-making and recommendation.
- **Structuring:** Logical approach of gathering and analysing information
- **Synthesis:** Approach towards interpreting information and making informed decisions
- **Business Judgement:** Ability to evaluate facts, and consider risks and possible implications when making decisions.

Key Performance Indicators (KPIs)

- The KPI offer insights into various aspects of logistics operations, including costs, revenue, overall effectiveness, and the fulfilment of Turnaround Time (TAT) obligations.

The Revenue Section

- The revenue section displays the amount of revenue generated from logistics operations utilizing various transportation methods.

Turnaround Time

- The Turnaround Time (TAT) section quantifies the variance between anticipated turnaround time and the actual delivery durations.

Inbound and Outbound Logistics Cost

- The inbound logistics cost encompasses the total expenses involved in bringing goods into the system as determined by the decisions made in inbound logistics.
- Outbound Logistics Cost includes expenses associated with sending goods out. Emissions data measures the environmental impact resulting from logistics operations.

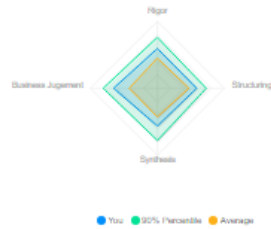
Round 1

XLS



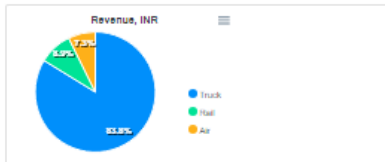
Report

Excellent work. You have completed optimizing the logistics cost for your transportation company. In this exercise every decision you have taken shows your skills in dimensions like rigor, synthesis, structuring and business judgement.



KPI	
Logistics Cost, INR	4,253,309
Revenue, INR	25,646,546
Effectiveness, %	17
TAT obligation met	No

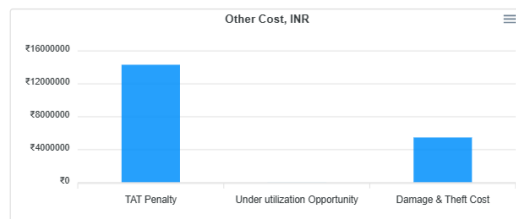
Revenue, INR	
Trucks	21,486,546
Rail	2,288,000
Air	1,872,000



Turnaround Time			
	Actual TAT	Difference %	Expected TAT
Unison Limited	8	-3	8
Prompton Incorporation	7	28	9
Fix Corporate	4	69	12
eCom Limited	6	-21	5



Inbound Logistics TAT, days	
Unison Limited	2.6
Prompton Incorporation	2.3
Fix Corporate	7.9
eCom Limited	12.3



Inbound Logistics Cost, INR	
Fuel Cost	45,346
Fleet Wages Cost	138,544
Contract Wage Cost	5,421
Warehouse Technology Cost	220,000
Under Utilization Opportunity Cost	23,100
Other Cost	530,203
Total Cost	962,615

Outbound Logistics Cost, INR	
Fuel Cost	1,480,477
Fleet Wages Cost	1,974,256
Contract Wage Cost	0
Technology Cost	300,000
Other costs	19,822,863
Total Cost	23,577,597

Figure 17 The report tab

2.9 Synopsis

The synopsis section summarizes the overall decision areas.

Round 1

XLS

Synopsis

Decisions	
Truck 1	30
Truck 2	20
Truck 3	30
Truck 1, Optimization	Yes
Truck 2, Optimization	eCom Limited
Truck 3, Optimization	Yes
Unison Limited	First-In-First-Out (FIFO)
Prompton Incorporation	First-In-First-Out (FIFO)
Fix Corporate	First-In-First-Out (FIFO)
eCom Limited	First-In-First-Out (FIFO)
Unison Limited, docks	6
Prompton Incorporation, docks	5
Fix Corporate, docks	4
eCom Limited, docks	3
Warehouse Upgrade 1	Conveyor System
Warehouse Upgrade 2	Stacker Cranes
Route	Delhi - Ahmedabad - Bangalore
Technology 1	GPS Positioning on Trucks
Technology 2	TAT Visibility
Small Trucks	40
Open Body Trucks	40
Covered Container	50
Rail Trips	4
Air Trips	3
Small Trucks, Optimization	Yes
Open Body Trucks, Optimization	Yes
Covered Container, Optimization	Yes
Rail, Optimization	Fix Corporate
Air, Optimization	eCom Limited

Inbound Logistics Cost, INR	
Fuel Cost	12,628
Fleet Wages Cost	24,800
Contract Wages Cost	6,319
Warehouse Technology Cost	510,000
Under Utilization Opportunity Cost	26,000
Other Cost	26,146,234
Total Cost	841,210

Inbound Logistics TAT, days	
Unison Limited	0.4
Prompton Incorporation	0.4
Fix Corporate	0
eCom Limited	0.2

Outbound Logistics Cost, INR	
Fuel Cost	1,500,193
Fleet Wages Cost	631,157
Contract Wages Cost	520,705
Technology Cost	610,000
Other costs	150,044
Total Cost	3,412,099

KPI	
Logistics Cost, INR	4,253,309
Revenue, INR	25,646,546
Effectiveness, %	17
TAT obligation met	No

Turnaround Time, days	
Unison Limited	8
Prompton Incorporation	7
Fix Corporate	4
eCom Limited	6

Revenue, INR	
Trucks	21,486,546
Rail	2,288,000
Air	1,872,000

Thinking Ability, %	
Inbound	50
Warehouse	0
Routes & Technology	50
Outbound	66.67

Round 2

Figure 18 The synopsis tab